

Feature audit — iOS · Android · Swiftcargo-main backend

Date: 2026-05-16 Repos:

- Backend: `~/Documents/PROJECTS/Swiftcargo-main` (Express + Postgres-on-Supabase, Railway-deployed; still the production backend until the Thapsus-webapp-and-backend cutover)
- Mobile: `~/Documents/PROJECTS/thapsus-v1.1` — `iosApp/` (SwiftUI, iOS 26 Liquid Glass) + `androidApp/` (Jetpack Compose)
- Shared KMP module owns 100% of business logic; this audit is purely about the **UI surface** that each client exposes vs the **HTTP surface** the backend offers.

Method: three parallel codebase enumerations (backend route files, iOS view files, Android Compose screens), then cross-referenced by domain.

TL;DR

- Feature scope is broadly at parity across iOS and Android.** The Android catch-up plan (Phases 0–5) is in, and the gap to iOS is now narrow.
- One real parity gap with a customer/admin impact:** Android `AdminPaymentsScreen` is still a Phase 0 stub. The route exists, but the screen renders a placeholder — admin payment review is blocked on Android until a Phase 5.3-style rewrite lands.
- A handful of smaller gaps** (label/manifest printing, AML risk queue surfacing, public unauth tracking, admin user detail drill-down, home greeting carousel) — listed in §4.
- Several backend domains are not consumed by either mobile client** — promotions/promo codes, admin backups, manual FX override, SSE event stream, admin email-config and test-email endpoints, the legacy `/api/wallet/*` (410 Gone) and `/api/consolidation V1` (deprecated). Most are intentionally web-only.
- One contradiction to verify in code:** the Android `NewOrderScreen` audit notes an "insurance tier" input on the form, but `project_insurance_removed.md` says the customer-facing picker was stripped on Android in PR #94 (2026-05-13). One of the two is wrong; flagged in §5.

1. Coverage matrix

Legend: ✓ = wired UI consumes the backend domain · **stub** = route + scaffold exist but UI is placeholder · — = no UI for that surface on this client · **n/a** = not applicable (web-only, internal, etc.)

#	Backend domain	Roles	iOS	Android	Notes
1	Auth — register/login/profile/password/reset/JWT exchange	All	✓	✓	iOS has explicit <code>ForgotPasswordView</code> ; Android folds it into <code>SignInScreen</code> + <code>PasswordResetScreen</code> . Functional parity.
2	Buy-for-me (concierge)	Customer + Ops + Admin	✓	✓	iOS embeds quote cards into Tracking; Android keeps quotes on Shop only (per scope-singular rule). Both deliberate.
3	Parcel orders (pre-register)	Customer + Admin	✓	✓	
4	Tracking (auth + public)	Customer + Public	✓	partial	iOS has <code>PublicTrackingViewModel</code> for unauthenticated THP-prefix tracking. Android has no public-tracking surface in audit. Gap.
5	Payments — Stripe + M-Pesa + Lipana STK	Customer	✓	✓	Both fully wired post PR #76 (Stripe SDK on Android). Comments in <code>PayInvoiceScreen</code> claiming "SDK not wired" are stale.
6	Public payment (deep-link <code>/pay/:id</code>)	Public	✓	✓	
7	Pending payment review	Admin	✓	stub	Android <code>AdminPaymentsScreen</code> is Phase 0 placeholder ("Check back soon"). Real gap.
8	Credit / referral wallet	Customer	✓	✓	Both use <code>CreditCenter</code> post-wallet-rip.
9	Transactions ledger	Customer	✓	✓	
10	Referral program	Customer	✓	✓	
11	Customer consolidation (self-service)	Customer + Admin	✓	✓	
12	Consolidations V2 (operator)	Operator + Admin	✓	✓	Shared <code>ConsolidationDetailScreen</code> between ops + admin on Android — clean reuse.
13	Invoices archive	Customer	✓	✓	Realtime via Supabase channel on both.
14	Issue invoice	Admin	✓	✓	
15	Pricing calculation / FX	Customer + Admin	partial	partial	Both have customer quote UIs. Neither has an admin UI for HS-codes / customs tiers / electronics fees / manual FX override —

#	Backend domain	Roles	iOS	Android	Notes
					managed via DB/SQL or web frontend. Probably fine.
16	Insurance policies + claims	Customer + Admin	n/a	n/a	Customer picker removed 2026-04-30 (web/iOS) and 2026-05-13 (Android). Backend routes remain operational for admin/historical use. Shared enum kept. See \$5 for one inconsistency.
17	Promo codes / pricing tiers	Customer + Admin	—	—	No mobile UI for promotion validation; backend domain unused by mobile. Probably web-only.
18	Warehouse intake (operator)	Operator	✓	✓	Both have scanner-driven receive flow.
19	Hardware printing (labels + manifests)	Operator	✓	partial	iOS has explicit <code>LabelPrinter</code> (<code>AirPrint</code>) + <code>ManifestPrinter</code> + <code>ReceiveLabelSheet</code> . Android: <code>androidx.print</code> declared and <code>ClientTerminalScreen</code> mentions print calls (per PR #79), but real scope and parity unclear — flagged.
20	Last-mile runs + POD (rider)	Rider + Operator (dispatch)	✓	✓	Android uses raw <code>LocationManager</code> + <code>URLConnection</code> upload (no-new-deps workaround); functionally equivalent.
21	Customs / clearing agent	Agent + Admin	✓	✓	
22	Agent invoices	Agent + Admin	✓	✓	
23	Tickets (support)	Customer + Admin	✓	✓	
24	NPS survey	Customer + Admin	✓	✓	iOS auto-prompt disabled per UX feedback; manual trigger via home carousel still works. Android exposes as bottom sheet.
25	DSAR (GDPR)	Customer + Admin	✓	✓	Both have customer form + admin queue.
26	AML flags	Admin	✓	—	iOS <code>AdminDashboardView</code> surfaces "AML risk queue"; Android <code>AdminConsoleScreen</code> doesn't expose one. Gap .
27	Admin user management	Admin	✓	partial	iOS has <code>AdminUsersView</code> + <code>AdminUserDetailView</code> (drill-down). Android audit shows <code>AdminUsersScreen</code> (list) but no detail screen — likely gap , worth verifying.
28	Admin orders	Admin	✓	✓	List + detail on both.
29	Admin revenue / KPI	Admin	✓	✓	iOS has separate <code>KPIDashboardView</code> ; Android has <code>KPIDashboardScreen</code> on its own tab. Backend revenue <code>UNIONS transactions + payments</code> — both clients consume the corrected endpoint.
30	Audit logs	Admin	✓	✓	
31	Error logs	Admin	✓	✓	
32	Backups	Admin	—	—	Backend <code>/api/admin/backups</code> not surfaced on either client. Admin DBA-only — intentional.
33	Email config check	Admin	partial	—	iOS <code>AdminDashboardView</code> shows Gmail readiness card (client ID lengths, sender email). Android omits. Minor gap.
34	Test email endpoint	Admin	—	—	Dev helper; neither client.
35	Notifications (in-app inbox)	Customer + Admin	✓	✓	
36	Notification banner (transient toast)	All	✓	✓	
37	Retailers directory	Customer	✓	✓	Marquee on iOS, catalogue on Android — both wired.
38	App config / feature flags	All	likely	likely	Both clients are expected to hit <code>/api/app-config</code> at boot; not explicitly enumerated in screen audits because it's plumbing, not a screen.
39	Warehouse addresses	Customer	✓	✓	
40	Prohibited items search	Customer	✓	✓	Dual-source (server canonical + bundled <code>ProhibitedItemsCatalog.kt</code> fallback) —

#	Backend domain	Roles	iOS	Android	Notes
					see <code>project_prohibited_items_catalogue.md</code> .
41	Exchange rates (read)	All	✓	✓	Quote calculators on both use server <code>/exchange/rates</code> .
42	SSE event stream (<code>/api/events</code>)	All authenticated	—	—	Neither mobile client consumes SSE; both use Supabase Realtime channels instead. Backend route is web-only.
43	Stripe webhook	n/a (server)	n/a	n/a	
44	Lipana M-Pesa webhook	n/a (server)	n/a	n/a	
45	Sitemap / robots / SEO	n/a	n/a	n/a	Web-only.
46	Legacy <code>/api/wallet/*</code>	—	—	—	Returns 410 Gone since mig 028. Correctly unused.
47	Legacy <code>/api/consolidation V1</code>	—	—	—	Sunset 2026-05-23. Correctly unused.
48	Home greeting carousel / pending actions	Customer	✓	partial	iOS has <code>HomeGreetingCarousel</code> + <code>HomeGreetingNavigation</code> + <code>PendingActionsView</code> . Android has a <code>HomeScreen</code> + <code>PendingActionsScreen</code> but no carousel mechanism. Likely Phase 6 polish item.
49	Appearance / theme settings	All	✓	✓	
50	Ops settings	Admin	✓	partial	Android <code>OpsSettingsScreen</code> exists but per audit "some fields may be TBD". Worth verifying.
51	Client terminal (walk-in kiosk)	Operator	✓ (<code>ClientTerminalView</code>)	✓ (<code>ClientTerminalScreen</code>)	Both wired. iOS labels it a "debug/test surface"; Android describes it as a real walk-in kiosk. Behavioural parity worth double-checking.

Totals: ~51 backend feature domains. iOS exposes ~43, Android exposes ~41 (with one stub). Most of the 8–10 backend-only domains are intentionally web/admin/internal-only.

2. iOS-only surfaces (not present on Android)

Surface	iOS file	Why it matters
Public unauthenticated tracking	<code>PublicTrackingViewModel</code> — reads THP via Supabase RLS without login	Customers can paste a tracking number into the app without signing in. Android offers no equivalent.
Admin user detail drill-down	<code>AdminUserDetailView</code>	Android has the user list but no detail screen — admin can't edit role/risk notes from a phone.
AML risk queue surfacing	<code>AdminDashboardView</code> AML section	Compliance touchpoint. Backend domain works; Android admin can't see it.
Email config readiness check	<code>AdminDashboardView</code> Gmail card	Minor — admin can verify SMTP without leaving the dashboard.
Home greeting carousel + auto-prompts	<code>HomeGreetingCarousel</code> , <code>HomeGreetingNavigation</code> , <code>NpsAutoPromptModifier</code>	Customer engagement loop. Android <code>HomeScreen</code> is static-section based.
Forgot-password as separate screen	<code>ForgotPasswordView</code>	UX preference — Android folds it into sign-in. Functional parity.
Explicit hardware label/manifest printing	<code>LabelPrinter</code> , <code>ManifestPrinter</code> , <code>ReceiveLabelSheet</code> , <code>SkuScannerView</code> (with print)	Android has <code>androidx.print</code> declared and <code>ClientTerminalScreen</code> calls into print per PR #79, but the dedicated "print a SKU label on receive" sheet flow needs verification.

3. Android-only surfaces (not present on iOS)

None of substance. The Android audit lists no domain that lacks an iOS counterpart. Where iOS is the design lead, Android trails; nothing flipped the other way.

(The Android audit notes "ML Kit barcode scanning for recipient ID on POD". iOS scanner uses native `AVFoundation` so this is implementation detail, not a missing feature.)

4. Backend-only surfaces (neither client consumes)

These are intentional in most cases — flagging so we have a complete picture:

- `/api/admin/backups` — DBA-only, no mobile UI ever planned.
- `/api/admin/test-email`, `/api/admin/email-config` — dev helpers. iOS surfaces config-readiness; neither surfaces the test-send.
- `/api/admin/exchange-rates` (**PUT manual override + refresh trigger**) — manual FX override is web/admin only.

- `/api/events` (SSE) — both mobile clients use Supabase Realtime instead. Either deprecate SSE or document it as web-only.
- `** /api/pricing/*` admin mutations (HS-codes, customs tiers, electronics fees, settings)** — managed via DB/SQL or web frontend.
- `** /api/pricing-tiers/*` (promotions, fees)** — promo codes have no mobile UI. If promos are part of the product, this is a gap; if web-only, fine.
- `/api/insurance/quote + policies + claims` — customer picker removed by product decision (2026-04-30 web/iOS, 2026-05-13 Android). Backend remains operational for admin/historical use only.
- `**Legacy /api/wallet/*` — 410 Gone since mig 028. Correctly unused.
- **Legacy** `/api/consolidation V1` — sunset 2026-05-23. Correctly unused.

5. Findings worth a follow-up

F1 — Android `AdminPaymentsScreen` is a stub (Phase 5.3 work)

Severity: real gap. Admin cannot approve pending M-Pesa payments from an Android device. iOS has the queue. This is the only post-Phase-5 feature gap noted in both audits. **Action:** Schedule a P5.4-equivalent PR. Surface `AdminApprovePaymentScreen` + integrate `/admin/payments/pending`, `/admin/payments/:id/approve`, `/admin/payments/:id/reject`.

F2 — Conflict: Android `NewOrderScreen` "insurance tier" input

The Android audit notes the form collects "insurance tier" as a field. `project_insurance_removed.md` says PR #94 stripped the customer picker on Android on 2026-05-13. Either the audit picked up a stale shared-VM parameter that isn't surfaced as a UI control (most likely — the shared `InsuranceTier` enum is kept and clients pass `STANDARD`), or the picker survived the cleanup. **Action:** read `androidApp/.../NewOrderScreen.kt` and confirm there is no segmented control / radio for insurance tier. If there is, remove it.

F3 — Android lacks an unauthenticated public-tracking surface

Severity: medium. iOS lets a customer paste a THP into a "track without signing in" search. Android only exposes tracking after login. Public tracking is a low-friction acquisition surface and the backend route (`/api/tracking/:trackingNumber`) supports it. **Action:** add a public-tracking screen on the Android sign-in page, mirroring iOS.

F4 — Android `AdminUsersScreen` has no detail drill-down

Severity: medium. The list exists, but per the audit there's no `AdminUserDetailScreen` to view/edit role, risk notes, AML flags, or to reset password/resend welcome. iOS has `AdminUserDetailView`. **Action:** verify in code (audit may have missed it), and if absent, add a P5-style PR.

F5 — Android admin scaffold doesn't expose AML queue

Severity: medium. iOS `AdminDashboardView` shows AML risk queue cards. Android `AdminConsoleScreen` doesn't enumerate one in the audit. Backend `/api/admin/aml-flags` is functional. **Action:** add an `AdminAmlQueueScreen` and link it from the console.

F6 — Android operator label/manifest printing parity

Severity: low-medium. iOS has explicit `ReceiveLabelSheet` + `LabelPrinter` AirPrint sheet on receive. Android: per PR #79, `ClientTerminalScreen` calls `LabelPrinter.print` and `ManifestPrinter.print`, but the receive-flow printing UX flow may differ. Worth a side-by-side run. **Action:** test on an Android device with a print service installed; compare to iOS receive flow.

F7 — Android home lacks the greeting carousel + auto-prompts

Severity: low (Phase 6 polish territory). iOS has a `HomeGreetingCarousel` driving customer re-engagement. Android `HomeScreen` shows static sections. **Action:** group with Phase 6 polish if/when scheduled.

F8 — SSE endpoint unused on mobile

Severity: informational. `/api/events` exists but neither client uses it (both use Supabase Realtime). If web is the only consumer, document it. If it's truly unused everywhere, schedule for deletion.

F9 — Stale comment in Android `PayInvoiceScreen`

The Android audit noted code comments claiming "Stripe SDK not yet wired into Gradle deps" — these contradict reality (SDK was added in PR #76 and `PaymentSheet` is live). Cosmetic cleanup, but the kind of stale claim that misleads future readers. **Action:** delete the comment when next touching that file.

F10 — Android `DispatchScreen` and `OpsSettingsScreen` have placeholder fields

Severity: low. The audit calls these "wired structure, scope TBD". Likely deliberate Phase 6 work, but worth listing the missing fields so we can size the work. **Action:** diff against iOS equivalents and produce a sub-issue.

6. Risk-ordered punch list

If we close these in order, Android reaches functional parity with iOS:

1. **F1** — Build the Android admin payment-approval screen. (P5.4)
2. **F4** — Add Android `AdminUserDetailScreen` (or confirm it exists and audit missed it).
3. **F5** — Add Android AML queue surface.
4. **F3** — Add Android public-tracking entry on sign-in.

5. **F2** — Verify and (if needed) remove residual insurance-tier input on `NewOrderScreen`.
6. **F6** — Verify Android receive-flow label printing end-to-end.
7. **F7, F10** — Group into Phase 6 (visual polish + small admin-screen gaps).
8. **F8, F9** — Documentation / cleanup, no user impact.

Item 1 is the only one with a customer-visible impact (an admin on Android can't approve a real customer's M-Pesa payment). Items 2–4 are operator/admin convenience parity. The rest are polish.

7. Methodology notes

- Three Explore agents ran in parallel — one per codebase — to keep the audit fresh and reduce cross-contamination.
- The auditors enumerated UI screens / route files only; they did NOT speculate about cross-platform parity (that synthesis is this report).
- Total surface counted: backend ~150 endpoints across 34 route files; iOS ~65 primary views; Android ~61 screens across 41 routes.
- Memory snapshots were used as context only; ground truth was the agents' direct reads of the current code on `main` of each repo.
- This audit is point-in-time. Feature parity should be re-checked after each phase of Android work lands.

Appendix — domain → route file → screen quick lookup

(Only the columns that have entries on all three sides — for cross-debugging.)

Domain	Backend file	iOS view	Android screen
Auth	<code>routes/auth.js</code>	<code>SignInView</code> , <code>PasswordResetView</code>	<code>SignInScreen</code> , <code>PasswordResetScreen</code>
BFM	<code>routes/buyForMe.js</code>	<code>BuyForMeView</code> , <code>OpsBuyForMeQueueView</code> , <code>AdminCreateBuyForMeView</code>	<code>BuyForMeScreen</code> , <code>OpsBuyForMeQueueScreen</code> , <code>AdminCreateBuyForMeScreen</code>
Orders	<code>routes/orders.js</code>	<code>NewOrderView</code> , <code>ParcelDetailView</code>	<code>NewOrderScreen</code> , <code>ParcelDetailScreen</code>
Tracking	<code>routes/tracking.js</code>	<code>TrackingView</code> , <code>PublicTrackingViewModel</code>	<code>TrackingScreen</code>
Payments	<code>routes/payments.js</code> , <code>payment.js</code>	<code>PayInvoiceView</code> , <code>MpesaSubmitSheet</code> , <code>LipanaStkSheet</code> , <code>PublicPaymentView</code>	<code>PayInvoiceScreen</code> , <code>MpesaSubmitBottomSheet</code> , <code>LipanaPhoneSheet</code> , <code>PublicPaymentScreen</code>
Payment review	<code>routes/adminPayments.js</code>	<code>AdminPaymentsView</code>	<code>AdminPaymentsScreen</code> (stub)
Credit	<code>routes/payments.js (/me/credit)</code>	<code>CreditCenterView</code> , <code>TransactionsView</code>	<code>CreditCenterScreen</code> , <code>TransactionsScreen</code>
Referral	<code>routes/referral.js</code>	<code>ReferralView</code>	<code>ReferralScreen</code>
Consolidation V2	<code>routes/consolidationsV2.js</code>	<code>ConsolidationListView</code> , <code>ConsolidationDetailView</code>	<code>ConsolidationListScreen</code> , <code>ConsolidationDetailScreen</code>
Customer consolidation	<code>routes/customerConsolidations.js</code>	<code>CustomerConsolidationView</code> , <code>CustomerInvoicesView</code>	<code>CustomerConsolidationsScreen</code> , <code>CustomerInvoicesScreen</code>
Last-mile	<code>routes/lastMile.js</code>	<code>RiderRunView</code> , <code>RunStopListView</code> , <code>PodCaptureView</code>	<code>RiderRunsScreen</code> , <code>RunStopListScreen</code> , <code>PodCaptureScreen</code>
Customs	<code>routes/customs.js</code>	<code>CustomsListView</code>	<code>CustomsScreen</code>
Agent invoices	<code>routes/agentInvoices.js</code>	<code>AgentInvoicesView</code>	<code>AgentInvoicesScreen</code>
Tickets	<code>routes/tickets.js</code>	<code>TicketsListView</code> , <code>TicketDetailView</code>	<code>TicketsScreen</code> , <code>TicketDetailScreen</code>
DSAR	<code>routes/dsar.js</code>	<code>DsarView</code> , <code>AdminDsarQueueView</code>	<code>DsarScreen</code> , <code>AdminDsarQueueScreen</code>
NPS	<code>routes/nps.js</code>	<code>NpsSurveyView</code>	<code>NpsSurveyBottomSheet</code>
Prohibited	<code>routes/prohibited.js</code>	<code>ProhibitedSearchView</code>	<code>ProhibitedSearchScreen</code>
Notifications	<code>routes/notifications.js</code>	<code>NotificationInboxView</code> , <code>NotificationBannerView</code>	<code>NotificationInboxScreen</code> , <code>NotificationBanner</code>
Warehouse addresses	<code>routes/warehouse.js</code>	<code>WarehouseAddressView</code>	<code>WarehouseAddressScreen</code>
Retailers	<code>routes/retailers.js</code>	retailer marquee inside <code>BuyForMeView</code>	retailer catalogue inside <code>BuyForMeScreen</code>
Exchange rates	<code>routes/exchange.js</code>	<code>QuoteCalculatorView</code>	<code>QuoteScreen</code>
Admin users	<code>routes/admin.js</code>	<code>AdminUsersView</code> , <code>AdminUserDetailView</code>	<code>AdminUsersScreen</code> (no detail per audit)
Admin orders	<code>routes/admin.js</code>	<code>AdminOrdersView</code> , <code>AdminOrderDetailView</code>	<code>AdminOrdersScreen</code> , <code>AdminOrderDetailScreen</code>
Admin revenue / KPI	<code>routes/admin.js</code>	<code>KPIDashboardView</code> , <code>AdminRevenueView</code>	<code>KPIDashboardScreen</code> , <code>AdminRevenueScreen</code>
Admin logs	<code>routes/admin.js</code>	<code>AdminAuditLogsView</code> , <code>AdminErrorLogsView</code>	<code>AdminAuditLogsScreen</code> , <code>AdminErrorLogsScreen</code>
AML flags	<code>routes/amlFlags.js</code>	<code>AdminDashboardView</code> (AML section)	(no screen)